50-DAY DATA ANALYST CHALLENGE · EXCEL DAY 03

Healthcare – Hospital Patient Service Analysis

Role: Junior Healthcare Data Analyst

BUSINESS SCENARIO

A multi-speciality hospital wants to improve appointment service. Management wants to identify waiting-time categories, cases requiring escalation, and appointments needing priority follow-up. This is an operational exercise; the sample contains no diagnoses or clinical decisions.

NEW CONCEPTS

IF, AND, OR, IFS and Nested IF. Translate written business rules into formulas, copy formulas down, and explain the logic.

BUSINESS RULES

- 15 minutes or less: Within Target.
- 16–30 minutes: Minor Delay.
- 31–45 minutes: Major Delay.
- More than 45 minutes: Critical Delay.
- Escalate if wait time is greater than 45 minutes OR feedback is Complaint.
- Priority Follow-up if wait time is greater than 30 minutes AND feedback is Complaint.

YOUR TASKS

1. Keep Appointment_Data unchanged. Create Service_Analysis and convert the data into an Excel Table named HospitalAppointments.

2. Add Wait_Category using IFS or Nested IF according to the supplied waiting-time rules.

3. Add Escalation_Status using IF with OR.

4. Add Follow_Up_Priority using IF with AND.

5. Create Business_Insights. Calculate total appointments and counts in each waiting-time category. COUNTIF may be used for counting; logical functions remain the main learning focus.

6. Identify the department with the highest average waiting time. Show your supporting figures.

7. How many appointments should be escalated? How many need priority follow-up? Are these groups identical? Explain.

8. Write two evidence-based observations and one operational recommendation.

SUBMISSION

Submit Day03_Hospital_Operations_Analysis.xlsx with Appointment_Data (unchanged), Service_Analysis and Business_Insights worksheets, plus README.md explaining formulas, findings, recommendation and assumptions.

GITHUB STRUCTURE

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50-Day-Data-Analyst-Challenge/Excel/Day-03/
Day03_Hospital_Operations_Analysis.xlsx
README.md
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Important: Work independently. The thresholds are exercise rules supplied by management, not medical triage standards.